Deep Learning Project Report

Equalization in Fiber-Optic Communication using Neural Networks

MICAS 913 – Deep Learning

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1 Part I: Simplified point-to-point communication system

1.1 Transmitter

Question 1

This question only requires implementing a Bernoulli source. The source was implemented as an i.i.d. binary process

$$b_i \sim \text{Bernoulli}(p), \qquad i = 0, 1, \ldots, N-1.$$

For the experiments in this project, $p = \frac{1}{2}$ was used, so zeros and ones are equally likely.

Question 2

A square 16-QAM constellation was generated and normalized so that the average symbol energy is

$$\frac{1}{M} \sum_{a \in C} |a|^2 = 1.$$

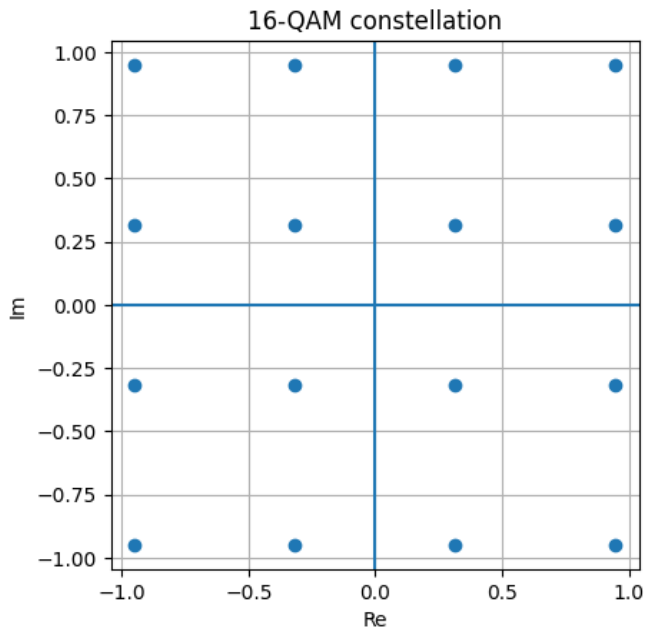


Figure 1: Normalized 16-QAM constellation.

```
Average energy = 0.9999999999999997
```

Question 3

The constellation is labeled using Gray coding. This ensures that nearest-neighbor points differ by only one bit, which reduces BER when symbol errors occur between adjacent points. The Gray mapping is constructed independently along the in-phase and quadrature axes and then concatenated.

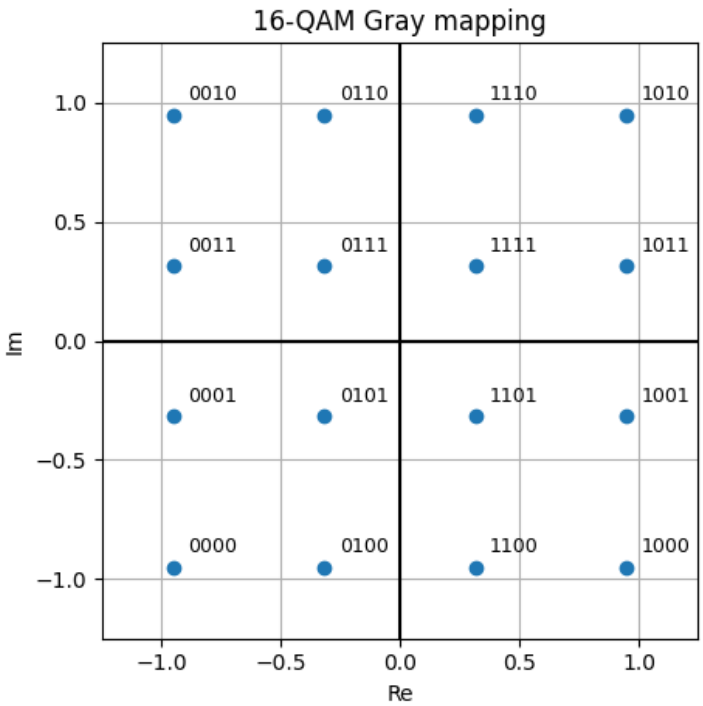


Figure 2: Gray labeling of the 16-QAM constellation.

Question 4

This question asks for the implementation of the bit-to-symbol mapper. The bit stream is partitioned into groups of $\log_2 M$ bits and each group is mapped to the corresponding Gray-labeled 16-QAM symbol:

$$b^N \longrightarrow s^{n_s}, \qquad n_s = \frac{N}{\log_2 M}.$$

```
bits = [1 1 0 0 1 0 0 0 1 1 1 0 0 0 0 1 0 0 1 1]
symbols = [ 0.31622777-0.9486833j   0.9486833 -0.9486833j   0.31622777+0.9486833j
 -0.9486833 -0.31622777j -0.9486833 +0.31622777j]
```

Question 5

Let T_s denote the symbol period. Then the symbol rate is

$$R_s = \frac{1}{T_s}.$$

For an M-QAM constellation, each symbol carries $\log_2 M$ bits, so the bit rate is

$$R_b = R_s \log_2 M.$$

This is the general formula.

Now consider the sinc-based signaling used in the guide. With the pulse

$$p(t) = \mathrm{sinc}(Bt),$$

the zero crossings occur at integer multiples of $1/B$. Therefore, the Nyquist symbol interval is

$$T_s = \frac{1}{B},$$

which implies the symbol rate

$$R_s = B.$$

Substituting this into the bit-rate formula gives

$$R_b = B \log_2 M.$$

For the specific case of 16-QAM,

$$R_b = 4B.$$

Question 6

Let

$$T_s = \frac{1}{B}$$

be the symbol period, and let $\alpha \in [0, 1)$ be the roll-off factor.

The root-raised-cosine (RRC) pulse has frequency response

$$P_{\mathrm{RRC}}(f) = \begin{cases} \sqrt{T_s}, & |f| \le \dfrac{1-\alpha}{2T_s}, \\ \sqrt{\dfrac{T_s}{2}\left[1 + \cos\left(\dfrac{\pi T_s}{\alpha}\left(|f| - \dfrac{1-\alpha}{2T_s}\right)\right)\right]}, & \dfrac{1-\alpha}{2T_s} < |f| \le \dfrac{1+\alpha}{2T_s}, \\ 0, & |f| > \dfrac{1+\alpha}{2T_s}. \end{cases}$$

Its time-domain expression is

$$p_{\mathrm{RRC}}(t) = \frac{1}{\sqrt{T_s}} \frac{\sin(\pi(1-\alpha)t/T_s) + 4\alpha(t/T_s)\cos(\pi(1+\alpha)t/T_s)}{\pi(t/T_s)\,(1 - (4\alpha t/T_s)^2)},$$

for the regular points $t \neq 0$ and $t \neq \pm\dfrac{T_s}{4\alpha}$.

At the singular points, the limiting values are

$$p_{\mathrm{RRC}}(0) = \frac{1}{\sqrt{T_s}}\left(1 + \alpha\left(\frac{4}{\pi} - 1\right)\right),$$

and

$$p_{\mathrm{RRC}}\left(\pm\frac{T_s}{4\alpha}\right) = \frac{\alpha}{\sqrt{2T_s}}\left[\left(1 + \frac{2}{\pi}\right)\sin\left(\frac{\pi}{4\alpha}\right) + \left(1 - \frac{2}{\pi}\right)\cos\left(\frac{\pi}{4\alpha}\right)\right].$$

In practice, the transmitter uses the RRC pulse and the receiver uses the matched RRC filter. The convolution of the two gives a raised-cosine response, which satisfies the Nyquist zero-ISI condition at symbol sampling instants.

Following the guide, we first validated the implementation using the sinc pulse shape. Once the transmitter and receiver chain were verified, we have also added the root-raised-cosine (RRC) pulse shape for the following experiments.

Question 7

The transmitted waveform is constructed by superposing shifted pulse shapes weighted by the transmitted symbols:

$$q(t, 0) = \sum_{\ell=\ell_1}^{\ell_2} s_\ell \, p(t - \ell T_s),$$

where $p(\cdot)$ is either the sinc pulse or the RRC pulse.

For the sinc case, using $T_s = 1/B$,

$$q(t,0) = \sum_{\ell=\ell_1}^{\ell_2} s_\ell \, \mathrm{sinc}(Bt - \ell).$$

In addition to the required plots, the pulse basis energy was checked numerically. For the adopted modulation convention, the pulse basis is expected to satisfy

$$\int_{-\infty}^{\infty} |p(t)|^2 \, dt \approx \frac{1}{B}.$$

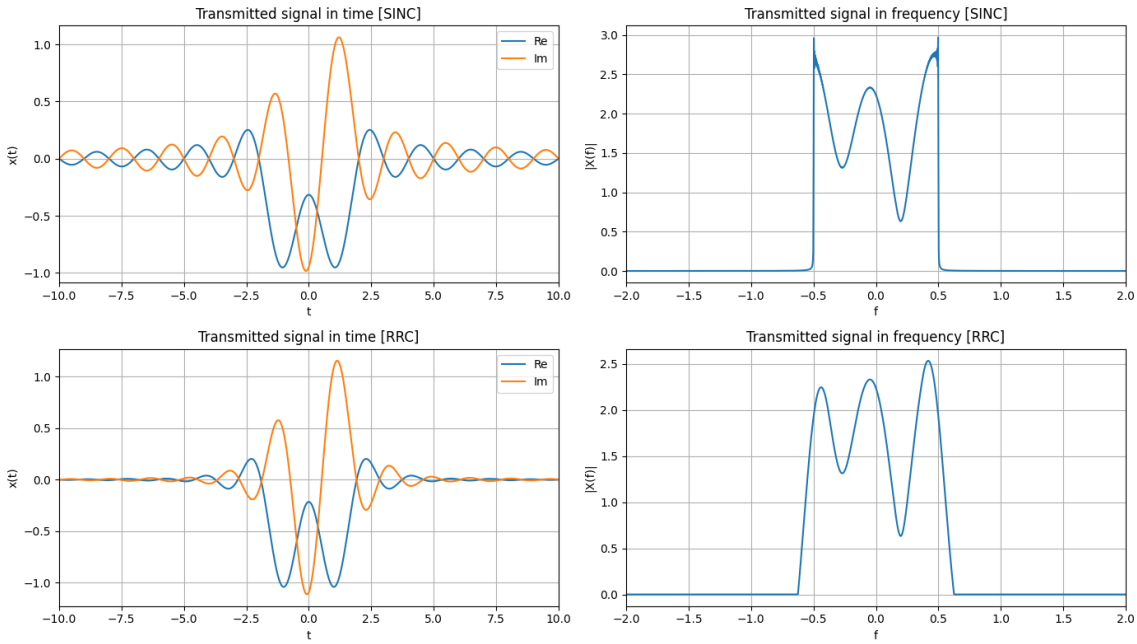


Figure 3: Transmitted signal in time and frequency domains for both pulse shapes: sinc and root-raised cosine (RRC).

```
Energy of basis pulse [SINC] = 0.99949339
Energy of basis pulse [RRC] = 1.00000000
```

1.2 Channel model

Question 8

Using the stochastic NLS:

$$j\frac{\partial Q(\tau,\ell)}{\partial \ell} = -\frac{\beta_2}{2}\frac{\partial^2 Q(\tau,\ell)}{\partial \tau^2} + \gamma |Q(\tau,\ell)|^2 Q(\tau,\ell) + N(\tau,\ell).$$

The left-hand side has unit

$$\left[\frac{\partial Q}{\partial \ell}\right] = \frac{\sqrt{\mathrm{W}}}{\mathrm{m}}.$$

For the dispersion term,

$$\left[\frac{\partial^2 Q}{\partial \tau^2}\right] = \frac{\sqrt{\mathrm{W}}}{\mathrm{s}^2},$$

so in order for

$$\beta_2 \frac{\partial^2 Q}{\partial \tau^2}$$

to have unit $\sqrt{\mathrm{W}}/\mathrm{m}$, we must have

$$[\beta_2] = \frac{\mathrm{s}^2}{\mathrm{m}}.$$

For the nonlinear term,

$$[|Q|^2 Q] = (\sqrt{\mathrm{W}})^3,$$

hence

$$[\gamma] = \frac{1}{\mathrm{W} \cdot \mathrm{m}} = \mathrm{W}^{-1}\mathrm{m}^{-1}.$$

For the noise term, the guide gives the correlation

$$\mathbb{E}[N(\tau,\ell)N^*(\tau',\ell')] = \sigma_0^2\, \delta_B(\tau-\tau')\, \delta(\ell-\ell').$$

Since $\delta_B(\tau-\tau')$ has unit 1/s and $\delta(\ell-\ell')$ has unit 1/m, and since the noise power formula is

$$P_n = \sigma_0^2 BL,$$

it follows that

$$[\sigma_0^2] = \frac{\mathrm{W}}{\mathrm{Hz} \cdot \mathrm{m}}.$$

Therefore,

$$[\beta_2] = \mathrm{s}^2/\mathrm{m}, \qquad [\gamma] = \mathrm{W}^{-1}\mathrm{m}^{-1}, \qquad [\sigma_0^2] = \mathrm{W}/(\mathrm{Hz} \cdot \mathrm{m}).$$

Question 9

$$q(t,z) = \frac{Q(\tau,\ell)}{\sqrt{P_0}}, \qquad t = \frac{\tau}{T_0}, \qquad z = \frac{\ell}{L}.$$

Equivalently,

$$Q(\tau,\ell) = \sqrt{P_0}\, q(t,z), \qquad \tau = T_0 t, \qquad \ell = Lz.$$

Using the chain rule,

$$\frac{\partial Q}{\partial \ell} = \frac{\sqrt{P_0}}{L}\frac{\partial q}{\partial z},$$

and

$$\frac{\partial^2 Q}{\partial \tau^2} = \frac{\sqrt{P_0}}{T_0^2}\frac{\partial^2 q}{\partial t^2}.$$

Also,

$$|Q|^2 Q = P_0^{3/2}|q|^2 q.$$

Substituting these expressions into the physical NLS equation gives

$$j\frac{\sqrt{P_0}}{L}\frac{\partial q}{\partial z} = -\frac{\beta_2}{2}\frac{\sqrt{P_0}}{T_0^2}\frac{\partial^2 q}{\partial t^2} + \gamma P_0^{3/2}|q|^2 q + N.$$

Dividing by $\sqrt{P_0}/L$, we obtain

$$j\frac{\partial q}{\partial z} = -\frac{\beta_2 L}{2T_0^2}\frac{\partial^2 q}{\partial t^2} + \gamma L P_0 |q|^2 q + n(t,z),$$

where $n(t,z)$ is the normalized noise term.

Now choose

$$P_0 = \frac{2}{\gamma L}, \qquad T_0 = \sqrt{\frac{|\beta_2| L}{2}}, \qquad L = L.$$

Since $\beta_2 < 0$,

$$-\frac{\beta_2 L}{2T_0^2} = 1,$$

and also

$$\gamma L P_0 = 2.$$

Therefore the normalized equation becomes

$$j\frac{\partial q(t,z)}{\partial z} = \frac{\partial^2 q(t,z)}{\partial t^2} + 2|q(t,z)|^2 q(t,z) + n(t,z).$$

The normalized noise correlation is written as

$$\mathbb{E}\big[n(t,z)n^*(t',z')\big] = \sigma^2 \delta_{B_n}(t-t')\delta(z-z'),$$

with

$$B_n = BT_0,$$

and

$$\sigma^2 = \frac{\sigma_0^2 L}{P_0 T_0}.$$

So the normalization removes the explicit physical parameters β_2 and γ, and the model is rewritten in dimensionless form.

Question 10

The fiber length is

$$L = 1000 \text{ km} = 10^6 \text{ m}.$$

The loss coefficient is given by $a_{\text{dB}} = 0.2$ dB/km. The guide gives

$$\alpha = 10^{-4}\log(10)\, a_{\text{dB}},$$

hence

$$\alpha = 10^{-4}\log(10) \times 0.2 = 4.60517 \times 10^{-5} \text{ m}^{-1}.$$

The dispersion parameter is

$$D = 17 \text{ ps/(nm} \cdot \text{km)}.$$

Converting to SI units,

$$D = \frac{17 \times 10^{-12}}{10^{-9} \times 10^3} = 17 \times 10^{-6} \text{ s/m}^2.$$

The carrier wavelength is

$$\lambda_0 = 1.55\ \mu\text{m} = 1.55 \times 10^{-6} \text{ m}.$$

The carrier frequency is

$$f_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{1.55 \times 10^{-6}} = 1.93548 \times 10^{14} \text{ Hz}.$$

The nonlinearity coefficient is

$$\gamma = 1.27 \text{ W}^{-1}\text{km}^{-1} = 1.27 \times 10^{-3} \text{ W}^{-1}\text{m}^{-1}.$$

The dispersion coefficient is obtained from

$$\beta_2 = -\frac{\lambda_0^2}{2\pi c} D.$$

Substituting the values,

$$\beta_2 = -\frac{(1.55 \times 10^{-6})^2}{2\pi(3 \times 10^8)}(17 \times 10^{-6}) = -2.16676 \times 10^{-26}\ \mathrm{s}^2/\mathrm{m}.$$

The scale factors are

$$L_0 = L = 10^6\ \mathrm{m},$$

$$T_0 = \sqrt{\frac{|\beta_2|L}{2}} = \sqrt{\frac{(2.16676 \times 10^{-26})(10^6)}{2}} = 1.04086 \times 10^{-10}\ \mathrm{s},$$

and

$$P_0 = \frac{2}{\gamma L} = \frac{2}{(1.27 \times 10^{-3})(10^6)} = 1.57480 \times 10^{-3}\ \mathrm{W}.$$

The physical noise PSD is

$$\sigma_0^2 = n_{\mathrm{sp}} h \alpha f_0.$$

With $n_{\mathrm{sp}} = 1$ and $h = 6.626 \times 10^{-34}\ \mathrm{J} \cdot \mathrm{s}$,

$$\sigma_0^2 = (1)(6.626 \times 10^{-34})(4.60517 \times 10^{-5})(1.93548 \times 10^{14}) = 5.90591 \times 10^{-24}\ \mathrm{W}/(\mathrm{Hz} \cdot \mathrm{m}).$$

The normalized noise variance is

$$\sigma^2 = \frac{\sigma_0^2 L}{P_0 T_0} = \frac{(5.90591 \times 10^{-24})(10^6)}{(1.57480 \times 10^{-3})(1.04086 \times 10^{-10})} = 3.60305 \times 10^{-5}.$$

Finally, the normalized bandwidth is

$$B_n = B T_0.$$

The code computes the same quantities numerically and prints

```
L       = 1000000.0
B       = 10000000000.0
f0      = 193548387096774.2
alpha   = 4.605170185988092e-05
beta2   = -2.1667619210769113e-26
L0      = 1000000.0
T0      = 1.0408558788508886e-10
P0      = 0.0015748031496062992
sigma02 = 5.905907932714276e-24
sigma2  = 3.603045929292216e-05
Bn      = 1.0408558788508886
```

Question 11

Normalized NLS equation in the absence of nonlinearity:

$$j\frac{\partial q(t,z)}{\partial z} = \frac{\partial^2 q(t,z)}{\partial t^2} + n(t,z).$$

The guide defines the Fourier transform with respect to t by

$$\hat{q}(\omega, z) = \int_{-\infty}^{\infty} q(t,z) e^{-j\omega t}\, dt$$

Using the Fourier transform property

$$\mathcal{F}\left\{\frac{\partial^2 q}{\partial t^2}\right\} = -\omega^2 \hat{q}(\omega, z),$$

we obtain

$$j\frac{\partial \hat{q}(\omega, z)}{\partial z} = -\omega^2 \hat{q}(\omega, z) + \hat{n}(\omega, z).$$

Rewriting,

$$\frac{\partial \hat{q}(\omega, z)}{\partial z} - j\omega^2 \hat{q}(\omega, z) = -j\hat{n}(\omega, z).$$

This is a first-order linear differential equation in z.
Multiplying by the integrating factor $e^{-j\omega^2 z}$ gives

$$\frac{\partial}{\partial z}\left(e^{-j\omega^2 z}\hat{q}(\omega, z)\right) = -je^{-j\omega^2 z}\hat{n}(\omega, z).$$

Integrating from 0 to z yields

$$e^{-j\omega^2 z}\hat{q}(\omega, z) - \hat{q}(\omega, 0) = -j\int_0^z e^{-j\omega^2 \zeta}\hat{n}(\omega, \zeta)\, d\zeta.$$

Multiplying both sides by $e^{j\omega^2 z}$, we obtain

$$\hat{q}(\omega, z) = e^{j\omega^2 z}\hat{q}(\omega, 0) - j\int_0^z e^{j\omega^2 (z-\zeta)}\hat{n}(\omega, \zeta)\, d\zeta.$$

Therefore the channel can be written in the required form

$$\hat{q}(\omega, z) = \hat{h}(\omega, z)\hat{q}(\omega, 0) + \hat{z}(\omega),$$

where

$$\hat{h}(\omega, z) = e^{j\omega^2 z},$$

and, for a fixed propagation distance z,

$$\hat{z}(\omega) = -j\int_0^z e^{j\omega^2 (z-\zeta)}\hat{n}(\omega, \zeta)\, d\zeta.$$

So the problem is to characterize the stochastic process $\hat{z}(\omega)$.

Mean of $\hat{z}(\omega)$. Since $n(t, z)$ is zero-mean circularly symmetric complex Gaussian noise, its Fourier transform $\hat{n}(\omega, z)$ is also zero mean. Therefore,

$$\mathbb{E}[\hat{z}(\omega)] = -j\int_0^z e^{j\omega^2 (z-\zeta)}\mathbb{E}[\hat{n}(\omega, \zeta)]\, d\zeta = 0.$$

Hence,

$$\mathbb{E}[\hat{z}(\omega)] = 0.$$

Correlation of $\hat{n}(\omega, z)$. From the guide, the normalized noise satisfies

$$\mathbb{E}[n(t,z)n^*(t',z')] = \sigma^2 \delta_{B_n}(t-t')\delta(z-z'),$$

where $B_n = BT_0$.

We now take the Fourier transform with respect to t:

$$\mathbb{E}[\hat{n}(\omega,z)\hat{n}^*(\omega',z')] = \iint \mathbb{E}[n(t,z)n^*(t',z')]\, e^{-j\omega t} e^{j\omega' t'}\ dt\ dt'.$$

Substituting the autocorrelation,

$$\mathbb{E}[\hat{n}(\omega,z)\hat{n}^*(\omega',z')] = \sigma^2 \delta(z-z') \iint \delta_{B_n}(t-t') e^{-j\omega t} e^{j\omega' t'}\ dt\ dt'.$$

The Fourier transform of $\delta_{B_n}(t-t')$ is the indicator of the band $|\omega| \le \pi B_n$. Therefore,

$$\mathbb{E}[\hat{n}(\omega,z)\hat{n}^*(\omega',z')] = 2\pi\sigma^2 \mathbf{1}_{\{|\omega|\le \pi B_n\}} \delta(\omega-\omega')\delta(z-z').$$

Thus the frequency-domain noise is white in ω inside the band.

Correlation of $\hat{z}(\omega)$. Using the expression of $\hat{z}(\omega)$,

$$\hat{z}(\omega) = -j \int_0^z e^{j\omega^2(z-\zeta)} \hat{n}(\omega,\zeta)\ d\zeta,$$

we compute

$$\mathbb{E}[\hat{z}(\omega)\hat{z}^*(\omega')] = \int_0^z \int_0^z e^{j\omega^2(z-\zeta)} e^{-j\omega'^2(z-\zeta')} \mathbb{E}[\hat{n}(\omega,\zeta)\hat{n}^*(\omega',\zeta')]\ d\zeta\ d\zeta'.$$

Substituting the correlation of $\hat{n}$,

$$\mathbb{E}[\hat{z}(\omega)\hat{z}^*(\omega')] = 2\pi\sigma^2 \mathbf{1}_{\{|\omega|\le \pi B_n\}} \delta(\omega-\omega') \int_0^z \int_0^z e^{j\omega^2(z-\zeta)} e^{-j\omega'^2(z-\zeta')} \delta(\zeta-\zeta')\, d\zeta\ d\zeta'.$$

The Dirac delta $\delta(\zeta-\zeta')$ collapses one integral, so

$$\mathbb{E}[\hat{z}(\omega)\hat{z}^*(\omega')] = 2\pi\sigma^2 \mathbf{1}_{\{|\omega|\le \pi B_n\}} \delta(\omega-\omega') \int_0^z e^{j\omega^2(z-\zeta)} e^{-j\omega'^2(z-\zeta)}\ d\zeta.$$

Because of the factor $\delta(\omega-\omega')$, we can set $\omega' = \omega$ inside the integral. The exponential terms cancel, giving

$$\mathbb{E}[\hat{z}(\omega)\hat{z}^*(\omega')] = 2\pi\sigma^2 z\, \mathbf{1}_{\{|\omega|\le \pi B_n\}} \delta(\omega-\omega').$$

Therefore, the correlation function of $\hat{z}(\omega)$ is

$$\mathbb{E}[\hat{z}(\omega)\hat{z}^*(\omega')] = 2\pi\sigma^2 z\, \mathbf{1}_{\{|\omega|\le \pi B_n\}} \delta(\omega-\omega').$$

This shows that $\hat{z}(\omega)$ is a white circularly symmetric complex Gaussian process in frequency over the noise band, and its variance grows linearly with the propagation distance z.

PDF of $\hat{z}(\omega)$. Since $\hat{z}(\omega)$ is a linear transform of the Gaussian noise $n(t,z)$, it is also a circularly symmetric complex Gaussian process.

In continuous frequency, the most precise characterization is therefore through its mean and correlation function rather than a pointwise scalar PDF.

For a fixed propagation distance z, $\hat{z}(\omega)$ is a zero-mean circularly symmetric complex Gaussian process satisfying

$$\mathbb{E}[\hat{z}(\omega)] = 0,$$

and

$$\mathbb{E}[\hat{z}(\omega)\hat{z}^*(\omega')] = 2\pi\sigma^2 z\, \mathbf{1}_{\{|\omega|\le \pi B_n\}} \delta(\omega-\omega').$$

Thus, $\hat{z}(\omega)$ is white in frequency over the band $|\omega| \le \pi B_n$, and its power grows linearly with the propagation distance z.

After discretization on the frequency mesh, the in-band frequency samples become complex Gaussian random variables.

Final answer. For a fixed propagation distance z, the linearized channel in frequency is

$$\hat{q}(\omega, z) = e^{j\omega^2 z}\hat{q}(\omega, 0) + \hat{z}(\omega),$$

where $\hat{z}(\omega)$ is a zero-mean circularly symmetric complex Gaussian process.

Its mean is

$$\mathbb{E}[\hat{z}(\omega)] = 0,$$

and its autocorrelation function is

$$\mathbb{E}\big[\hat{z}(\omega)\hat{z}^*(\omega')\big] = 2\pi\sigma^2 z\, \mathbf{1}_{\{|\omega|\le \pi B_n\}}\delta(\omega - \omega').$$

Therefore, $\hat{z}(\omega)$ is white in frequency over the band $|\omega| \le \pi B_n$, and its power grows linearly with the propagation distance z.

After discretization on the frequency mesh, the in-band frequency samples become circularly symmetric complex Gaussian random variables.

Question 12

Linearized channel in the frequency domain is

$$\hat{q}(\omega, z) = \hat{h}(\omega, z)\hat{q}(\omega, 0) + \hat{z}(\omega),$$

with

$$\hat{h}(\omega, z) = e^{j\omega^2 z}.$$

Taking the inverse Fourier transform, we obtain the time-domain channel

$$q(t, z) = h(t, z) \star q(t, 0) + z(t),$$

where

$$h(t, z) = \mathcal{F}^{-1}\{\hat{h}(\omega, z)\}(t)$$

and

$$z(t) = \mathcal{F}^{-1}\{\hat{z}(\omega)\}(t).$$

Channel impulse response $h(t, z)$. Using the inverse Fourier transform convention,

$$h(t, z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{j\omega^2 z} e^{j\omega t}\, d\omega.$$

We complete the square in the exponent:

$$z\omega^2 + t\omega = z\left(\omega + \frac{t}{2z}\right)^2 - \frac{t^2}{4z}.$$

Hence

$$h(t, z) = \frac{e^{-jt^2/(4z)}}{2\pi}\int_{-\infty}^{\infty} e^{jz\left(\omega + \frac{t}{2z}\right)^2}\, d\omega.$$

With the change of variable

$$u = \omega + \frac{t}{2z},$$

this becomes

$$h(t, z) = \frac{e^{-jt^2/(4z)}}{2\pi}\int_{-\infty}^{\infty} e^{jzu^2}\, du.$$

We know

$$\int_{-\infty}^{\infty} e^{jzu^2}\, du = \sqrt{\frac{\pi}{z}}\, e^{j\pi/4}, \qquad z > 0,$$

so we get

$$h(t,z) = \frac{e^{j\pi/4}}{\sqrt{4\pi z}} \exp\left(-j\frac{t^2}{4z}\right), \qquad z > 0.$$

Therefore, the channel impulse response is

$$h(t,z) = \frac{e^{j\pi/4}}{\sqrt{4\pi z}} \exp\left(-j\frac{t^2}{4z}\right).$$

PDF of $z(t)$. From Question 11, $\hat{z}(\omega)$ is a zero-mean circularly symmetric complex Gaussian process. Since $z(t)$ is obtained from $\hat{z}(\omega)$ by an inverse Fourier transform, which is linear, $z(t)$ is also a zero-mean circularly symmetric complex Gaussian process.

Therefore, the PDF of $z(t)$ is Gaussian in the complex sense. More precisely, for each fixed time t,

$$z(t) \sim \mathcal{CN}(0, \sigma^2 z B_n).$$

So $z(t)$ is a bandlimited white circularly symmetric complex Gaussian noise process in time.

Mean of $z(t)$. Since $\hat{z}(\omega)$ has zero mean, its inverse Fourier transform also has zero mean:

$$\mathbb{E}[z(t)] = \mathcal{F}^{-1}\{\mathbb{E}[\hat{z}(\omega)]\}(t) = 0.$$

Hence,

$$\mathbb{E}[z(t)] = 0.$$

Correlation of $z(t)$. From Question 11, the correlation of $\hat{z}(\omega)$ is

$$\mathbb{E}\big[\hat{z}(\omega)\hat{z}^*(\omega')\big] = 2\pi\sigma^2 z\, \mathbf{1}_{\{|\omega|\le \pi B_n\}}\delta(\omega-\omega').$$

Now compute the time-domain correlation:

$$\mathbb{E}\big[z(t)z^*(t')\big] = \frac{1}{(2\pi)^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbb{E}\big[\hat{z}(\omega)\hat{z}^*(\omega')\big]\, e^{j\omega t}e^{-j\omega' t'}\, d\omega\, d\omega'.$$

Substituting the expression above,

$$\mathbb{E}\big[z(t)z^*(t')\big] = \frac{1}{(2\pi)^2}\iint 2\pi\sigma^2 z\, \mathbf{1}_{\{|\omega|\le \pi B_n\}}\delta(\omega-\omega')e^{j\omega t}e^{-j\omega' t'}\, d\omega\, d\omega'.$$

Using the Dirac delta $\delta(\omega-\omega')$, we obtain

$$\mathbb{E}\big[z(t)z^*(t')\big] = \frac{\sigma^2 z}{2\pi}\int_{-\pi B_n}^{\pi B_n} e^{j\omega(t-t')}\, d\omega.$$

Evaluating the integral,

$$\mathbb{E}\big[z(t)z^*(t')\big] = \sigma^2 z B_n\, \mathrm{sinc}\big(B_n(t-t')\big).$$

Since the guide defines

$$\delta_{B_n}(x) = B_n\, \mathrm{sinc}(B_n x),$$

we can write the result as

$$\mathbb{E}\big[z(t)z^*(t')\big] = \sigma^2 z\, \delta_{B_n}(t-t').$$

Therefore, $z(t)$ is a zero-mean bandlimited white circularly symmetric complex Gaussian process with autocorrelation

$$\mathbb{E}\big[z(t)z^*(t')\big] = \sigma^2 z\, \delta_{B_n}(t-t').$$

Final answer. The time-domain channel is

$$q(t,z) = h(t,z) \star q(t,0) + z(t),$$

with impulse response

$$h(t,z) = \frac{e^{j\pi/4}}{\sqrt{4\pi z}} \exp\left(-j\frac{t^2}{4z}\right).$$

The additive term $z(t)$ is a zero-mean circularly symmetric complex Gaussian process. Its statistics are

$$\mathbb{E}[z(t)] = 0,$$

and

$$\mathbb{E}\big[z(t)z^*(t')\big] = \sigma^2 z\, \delta_{B_n}(t-t').$$

Equivalently, for each fixed t,

$$z(t) \sim \mathcal{CN}(0, \sigma^2 z B_n).$$

Question 13

The continuous-time input-power constraint is

$$P = \lim_{T\to\infty} \frac{1}{T}\mathbb{E}\left[\int_{-T/2}^{T/2} |q(t,0)|^2\, dt\right].$$

To discretize it, we sample the interval uniformly with N points and step

$$\Delta t = \frac{T}{N}.$$

Let

$$t_i = -\frac{T}{2} + i\Delta t, \qquad i = 0, 1, \ldots, N-1,$$

and define the samples

$$q_i(0) = q(t_i, 0).$$

Approximating the integral by a Riemann sum gives

$$\int_{-T/2}^{T/2} |q(t,0)|^2\, dt \approx \sum_{i=0}^{N-1} |q_i(0)|^2 \Delta t.$$

Substituting this into the power expression,

$$P \approx \frac{1}{T}\mathbb{E}\left[\sum_{i=0}^{N-1} |q_i(0)|^2 \Delta t\right].$$

Since $\Delta t = T/N$, we get

$$P \approx \frac{1}{T}\cdot\frac{T}{N}\sum_{i=0}^{N-1} \mathbb{E}|q_i(0)|^2 = \frac{1}{N}\sum_{i=0}^{N-1} \mathbb{E}|q_i(0)|^2.$$

Therefore, in the limit $N \to \infty$,

$$P = \lim_{N\to\infty} \frac{1}{N}\sum_{i=0}^{N-1} \mathbb{E}|q_i(0)|^2.$$

So the discrete-time version of the input-power constraint is

$$\frac{1}{N}\mathbb{E}\|\mathbf{q}(0)\|_2^2 \approx P,$$

where

$$\mathbf{q}(0) = \big(q_0(0), q_1(0), \ldots, q_{N-1}(0)\big)^T.$$

Question 14

Let

$$t_n = -\frac{T}{2} + n\Delta t, \qquad n = 0, 1, \ldots, N-1, \qquad \Delta t = \frac{T}{N}.$$

The corresponding frequency mesh is

$$f_k = -\frac{F}{2} + k\Delta f, \qquad k = 0, 1, \ldots, N-1,$$

where

$$F = \frac{1}{\Delta t}, \qquad \Delta f = \frac{1}{T}.$$

In angular frequency,

$$\omega_k = 2\pi f_k, \qquad \Delta\omega = 2\pi\Delta f = \frac{2\pi}{T}.$$

1. Discretization of the Fourier integrals. The continuous Fourier transform is

$$\hat{x}(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, dt.$$

Sampling in time and approximating the integral by a Riemann sum gives

$$\hat{x}(\omega_k) \approx \sum_{n=0}^{N-1} x(t_n)e^{-j\omega_k t_n}\Delta t.$$

If we define

$$x_n = x(t_n),$$

then this becomes

$$\hat{x}(\omega_k) \approx \Delta t \sum_{n=0}^{N-1} x_n e^{-j\omega_k t_n}.$$

So we obtain

$$\hat{x}(\omega_k) \approx \Delta t\, \hat{x}_k,$$

where

$$\hat{x}_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi nk/N}$$

is the DFT of the sampled signal.

Now consider the inverse Fourier transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{x}(\omega)e^{j\omega t}\, d\omega.$$

Approximating the integral by a Riemann sum on the frequency mesh gives

$$x(t_n) \approx \frac{1}{2\pi}\sum_{k=0}^{N-1} \hat{x}(\omega_k)e^{j\omega_k t_n}\Delta\omega.$$

Since

$$\Delta\omega = \frac{2\pi}{T},$$

we get

$$x_n \approx \frac{1}{T}\sum_{k=0}^{N-1} \hat{x}(\omega_k)e^{j2\pi nk/N}.$$

Using $\hat{x}(\omega_k) \approx \Delta t\, \hat{x}_k$ and $T = N\Delta t$, we get

$$x_n \approx \frac{1}{N} \sum_{k=0}^{N-1} \hat{x}_k e^{j2\pi nk/N}.$$

Thus, the Fourier integrals reduce to the DFT/IDFT pair

$$\hat{x}(\omega_k) \approx \Delta t\, \hat{x}_k, \qquad x_n \approx \frac{1}{N} \sum_{k=0}^{N-1} \hat{x}_k e^{j2\pi nk/N}.$$

2. Approximation of the channel in the frequency domain. From Question 11, the linearized channel in frequency is

$$\hat{q}(\omega, z) = \hat{h}(\omega, z)\hat{q}(\omega, 0) + \hat{z}(\omega), \qquad \hat{h}(\omega, z) = e^{j\omega^2 z}.$$

Sampling this expression at $\omega = \omega_k$ and using

$$\hat{q}(\omega_k, z) \approx \Delta t\, \hat{q}_k(z),$$

we obtain

$$\Delta t\, \hat{q}_k(z) \approx e^{j\omega_k^2 z} \Delta t\, \hat{q}_k(0) + \hat{z}(\omega_k).$$

Dividing by Δt,

$$\hat{q}_k(z) \approx e^{j\omega_k^2 z} \hat{q}_k(0) + \hat{z}_k, \qquad \hat{z}_k = \frac{\hat{z}(\omega_k)}{\Delta t}.$$

Therefore, in vector form, the discrete frequency-domain channel is

$$\hat{\mathbf{q}}(z) = \Gamma(z)\hat{\mathbf{q}}(0) + \hat{z},$$

where

$$\Gamma(z) = \operatorname{diag}\left(e^{j\omega_0^2 z}, e^{j\omega_1^2 z}, \ldots, e^{j\omega_{N-1}^2 z}\right).$$

3. Time-domain channel model. Applying the inverse DFT, the discrete time-domain channel becomes

$$\mathbf{q}(z) = D^{-1}\Gamma(z)D\mathbf{q}(0) + z,$$

where D is the DFT matrix.

Equivalently,

$$\mathbf{q}(z) = W(z)\mathbf{q}(0) + z, \qquad W(z) = D^{-1}\Gamma(z)D.$$

4. Input constraints in time and frequency. From Question 13, the discrete-time input-power constraint is

$$\frac{1}{N}\mathbb{E}\|\mathbf{q}(0)\|_2^2 \approx P.$$

Using Parseval's identity for the DFT convention used here,

$$\sum_{n=0}^{N-1} |q_n(0)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |\hat{q}_k(0)|^2.$$

Therefore,

$$\frac{1}{N}\mathbb{E}\|\mathbf{q}(0)\|_2^2 = \frac{1}{N^2}\mathbb{E}\|\hat{\mathbf{q}}(0)\|_2^2.$$

Hence the equivalent frequency-domain constraint is

$$\frac{1}{N^2}\mathbb{E}\|\hat{\mathbf{q}}(0)\|_2^2 \approx P.$$

Final summary. The sampled frequency-domain channel is

$$\hat{\mathbf{q}}(z) = \Gamma(z)\hat{\mathbf{q}}(0) + \hat{z},$$

with

$$\Gamma_{k,k}(z) = e^{j\omega_k^2 z}.$$

The sampled time-domain channel is

$$\mathbf{q}(z) = W(z)\mathbf{q}(0) + z,$$

with

$$W(z) = D^{-1}\Gamma(z)D.$$

The input-power constraint is

$$\frac{1}{N}\mathbb{E}\|\mathbf{q}(0)\|_2^2 \approx P,$$

or equivalently,

$$\frac{1}{N^2}\mathbb{E}\|\hat{\mathbf{q}}(0)\|_2^2 \approx P.$$

Question 15

To discretize the noise, we sample the time-domain additive noise process on the mesh

$$t_0, \ldots, t_{N-1}$$

and obtain a vector

$$n(z) \in \mathbb{C}^N.$$

From the continuous-time autocorrelation

$$\mathbb{E}\big[n(t,z)n^*(t',z')\big] = \sigma^2\delta_B(t-t')\delta(z-z'),$$

the sampled discrete autocorrelation at a fixed distance z is

$$\mathbb{E}\Big[n_i(z)n_j^*(z)\Big] = \sigma^2 z\,\delta_B(t_i - t_j) = \sigma^2 z\,B\ \mathrm{sinc}(B(t_i - t_j)).$$

Hence the discrete noise covariance matrix is

$$\mathbb{E}\Big[n(z)n(z)^H\Big] = \Big[\sigma^2 zB\ \mathrm{sinc}(B(t_i - t_j))\Big]_{i,j}.$$

The variance of each sample is obtained by setting $i = j$:

$$\mathbb{E}|n_i(z)|^2 = \sigma^2 zB\,\mathrm{sinc}(0) = \sigma^2 Bz.$$

Rather than generating correlated Gaussian noise directly in time, we generate white Gaussian noise over the in-band frequency bins and then apply the inverse FFT. This produces a bandlimited Gaussian time-domain noise vector whose power matches the sampled continuous-time model.

Directly sampling the full correlated Gaussian vector in time would be more cumbersome, whereas frequency-domain sampling automatically enforces the bandlimitation.

1.3 Receiver

Question 16

From Question 10, before equalization the linear channel in frequency is

$$\hat{q}(\omega, L) = \hat{h}(\omega, L)\hat{q}(\omega, 0) + \hat{z}(\omega), \qquad \hat{h}(\omega, L) = e^{j\omega^2 L}.$$

The guide defines the equalized signal by

$$\hat{q}_e(\omega, L) = \hat{h}^{-1}(\omega, L)\hat{q}(\omega, L) = \hat{q}(\omega, 0) + \hat{w}(\omega),$$

where

$$\hat{w}(\omega) = \hat{h}^{-1}(\omega, L)\hat{z}(\omega).$$

In time domain,

$$q_e(t, L) = q(t, 0) + w(t),$$

where

$$w(t) = \mathcal{F}^{-1}\{\hat{w}(\omega)\}(t).$$

1. Characterization of $\hat{w}(\omega)$. Since $\hat{z}(\omega)$ is a zero-mean circularly symmetric complex Gaussian process and $\hat{w}(\omega)$ is obtained by multiplying it by the deterministic function $\hat{h}^{-1}(\omega, L)$, the process $\hat{w}(\omega)$ is also circularly symmetric complex Gaussian.

Its mean is

$$\mathbb{E}[\hat{w}(\omega)] = \hat{h}^{-1}(\omega, L)\mathbb{E}[\hat{z}(\omega)] = 0.$$

Now compute the autocorrelation:

$$\mathbb{E}\big[\hat{w}(\omega)\hat{w}^*(\omega')\big] = \hat{h}^{-1}(\omega, L)\hat{h}^{-1}(\omega', L)^*\mathbb{E}\big[\hat{z}(\omega)\hat{z}^*(\omega')\big].$$

From Question 10,

$$\mathbb{E}\big[\hat{z}(\omega)\hat{z}^*(\omega')\big] = 2\pi\sigma^2 L\, \mathbf{1}_{\{|\omega|\le \pi B_n\}}\delta(\omega - \omega').$$

Also,

$$\hat{h}^{-1}(\omega, L) = e^{-j\omega^2 L},$$

so

$$\hat{h}^{-1}(\omega, L)\hat{h}^{-1}(\omega', L)^* = e^{-j\omega^2 L}e^{j\omega'^2 L}.$$

Because of the factor $\delta(\omega - \omega')$, we set $\omega' = \omega$, and the phase terms cancel. Hence

$$\mathbb{E}\big[\hat{w}(\omega)\hat{w}^*(\omega')\big] = 2\pi\sigma^2 L\, \mathbf{1}_{\{|\omega|\le \pi B_n\}}\delta(\omega - \omega').$$

Therefore, $\hat{w}(\omega)$ has exactly the same second-order statistics as $\hat{z}(\omega)$.

So the frequency-domain equalization does not amplify the noise power, because the channel is all-pass:

$$|\hat{h}(\omega, L)| = 1.$$

2. PDF of $\hat{w}(\omega)$. As a stochastic process, $\hat{w}(\omega)$ is a zero-mean circularly symmetric complex Gaussian process in frequency.

Strictly speaking, in continuous frequency it is best characterized by its mean and correlation function above.

After discretization on the frequency mesh, each in-band frequency sample is complex Gaussian with

$$\hat{w}_k \sim \mathcal{CN}(0, 2\pi\sigma^2 L),$$

for the bins satisfying $|\omega_k| \leq \pi B_n$.

So the pointwise PDF of each in-band discrete coefficient is

$$p_{\hat{w}_k}(u) = \frac{1}{\pi(2\pi\sigma^2 L)} \exp\left(-\frac{|u|^2}{2\pi\sigma^2 L}\right).$$

Outside the noise band, the variance is zero.

3. Characterization of $w(t)$. Since

$$w(t) = \mathcal{F}^{-1}\{\hat{w}(\omega)\}(t)$$

and the inverse Fourier transform is linear, $w(t)$ is also a circularly symmetric complex Gaussian process.

Its mean is

$$\mathbb{E}[w(t)] = \mathcal{F}^{-1}\{\mathbb{E}[\hat{w}(\omega)]\}(t) = 0.$$

Its autocorrelation is the inverse Fourier transform of the frequency-domain autocorrelation. Therefore,

$$\mathbb{E}\big[w(t)w^*(t')\big] = \sigma^2 L\, \delta_{B_n}(t - t'),$$

where

$$\delta_{B_n}(x) = B_n \operatorname{sinc}(B_n x).$$

Thus, $w(t)$ is a zero-mean bandlimited white circularly symmetric complex Gaussian process in time.

Its variance is obtained by setting $t = t'$:

$$\mathbb{E}|w(t)|^2 = \sigma^2 L\, \delta_{B_n}(0) = \sigma^2 L B_n.$$

4. PDF of $w(t)$. For each fixed time t, the random variable $w(t)$ is circularly symmetric complex Gaussian with

$$w(t) \sim \mathcal{CN}(0, \sigma^2 L B_n).$$

Hence its PDF is

$$p_{w(t)}(u) = \frac{1}{\pi\sigma^2 L B_n} \exp\left(-\frac{|u|^2}{\sigma^2 L B_n}\right).$$

Final answer. The equalized noise in frequency,

$$\hat{w}(\omega) = \hat{h}^{-1}(\omega, L)\hat{z}(\omega),$$

is a zero-mean circularly symmetric complex Gaussian process with

$$\mathbb{E}[\hat{w}(\omega)] = 0,$$

and

$$\mathbb{E}\big[\hat{w}(\omega)\hat{w}^*(\omega')\big] = 2\pi\sigma^2 L\, \mathbf{1}_{\{|\omega| \leq \pi B_n\}} \delta(\omega - \omega').$$

So $\hat{w}(\omega)$ has the same statistics as $\hat{z}(\omega)$, because equalization multiplies by a unit-modulus phase factor only.

In time domain, the equalized noise $w(t)$ is also a zero-mean circularly symmetric complex Gaussian process with

$$\mathbb{E}[w(t)] = 0,$$

and

$$\mathbb{E}\big[w(t)w^*(t')\big] = \sigma^2 L\, \delta_{B_n}(t - t').$$

For each fixed time t,

$$w(t) \sim \mathcal{CN}(0, \sigma^2 L B_n).$$

Question 17

To validate the implementation, a Gaussian input signal was propagated through the noiseless channel and then equalized. Since the propagation is purely linear and invertible in this test, the equalized waveform should coincide with the original input.

The resulting plot confirms that:

- the signal broadens after propagation, as expected from dispersion,
- after equalization, the waveform is recovered with only negligible numerical error.

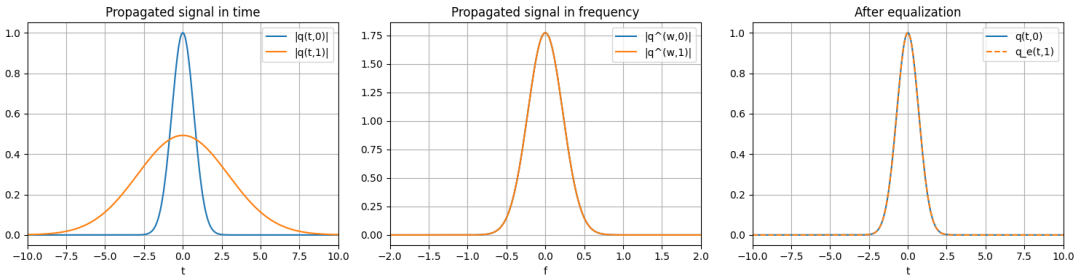


Figure 4: Validation of propagation and equalization on a Gaussian test signal.

```
max |q(t,0) - q_e(t,1)| = 6.6620e-16
```

Question 18

After equalization, the received symbol estimates are obtained by projecting the equalized waveform onto the pulse basis:

$$\hat{s}_\ell = B \int_{-\infty}^{\infty} q_e(t, z)\, p(t - \ell T_s)\; dt.$$

In practice, this integral is approximated numerically using the discrete time mesh:

$$\hat{s}_\ell \approx B\Delta t \sum_n q_e(t_n, z)\, p^*(t_n - \ell T_s).$$

The same pulse family used at the transmitter is used again at the receiver. Therefore:

- the sinc transmitter is matched with the sinc receiver,
- the RRC transmitter is matched with the RRC receiver.

Question 19

After equalization we have

$$q_e(t,z) = q(t,0) + w(t).$$

From the modulation model

$$q(t,0) = \sum_k s_k \operatorname{sinc}(Bt-k).$$

Substituting this into the demodulation formula

$$\hat{s}_\ell = B\int_{-\infty}^{\infty} q_e(t,z)\operatorname{sinc}(Bt-\ell)\ dt,$$

we obtain

$$\hat{s}_\ell = B\int_{-\infty}^{\infty}\left(\sum_k s_k \operatorname{sinc}(Bt-k) + w(t)\right)\operatorname{sinc}(Bt-\ell)\ dt.$$

By linearity,

$$\hat{s}_\ell = \sum_k s_k B\int_{-\infty}^{\infty}\operatorname{sinc}(Bt-k)\operatorname{sinc}(Bt-\ell)\ dt + B\int_{-\infty}^{\infty} w(t)\operatorname{sinc}(Bt-\ell)\ dt.$$

Now use the orthogonality of the sinc pulses:

$$B\int_{-\infty}^{\infty}\operatorname{sinc}(Bt-k)\operatorname{sinc}(Bt-\ell)\ dt = \delta_{k\ell}.$$

Therefore,

$$\hat{s}_\ell = s_\ell + Z_\ell,$$

where

$$Z_\ell = B\int_{-\infty}^{\infty} w(t)\operatorname{sinc}(Bt-\ell)\ dt.$$

So the continuous-time channel is reduced to the discrete-time channel

$$s^{n_s} \to \hat{s}^{n_s}, \qquad \hat{s}_\ell = s_\ell + Z_\ell.$$

Distribution of Z_ℓ. Since $w(t)$ is a zero-mean circularly symmetric complex Gaussian process, and Z_ℓ is a linear functional of $w(t)$, the random variable Z_ℓ is also circularly symmetric complex Gaussian.

Its mean is

$$\mathbb{E}[Z_\ell] = B\int_{-\infty}^{\infty}\mathbb{E}[w(t)]\operatorname{sinc}(Bt-\ell)\ dt = 0.$$

To compute the correlation, use the autocorrelation of the equalized noise from Question 16:

$$\mathbb{E}\big[w(t)w^*(t')\big] = \sigma^2 z\,\delta_B(t-t'), \qquad \delta_B(x) = B\operatorname{sinc}(Bx).$$

Then

$$\mathbb{E}[Z_\ell Z_m^*] = B^2\iint \mathbb{E}\big[w(t)w^*(t')\big]\operatorname{sinc}(Bt-\ell)\operatorname{sinc}(Bt'-m)\ dt\ dt'.$$

Substituting the correlation of $w(t)$,

$$E[Z_\ell Z_m^*] = \sigma^2 z B^2\int\int \delta_B(t-t')\operatorname{sinc}(Bt-\ell)\operatorname{sinc}(Bt'-m)\,dt\,dt'.$$

Since δ_B acts as the reproducing kernel for bandlimited signals,

$$\int_{-\infty}^{\infty}\delta_B(t-t')\operatorname{sinc}(Bt'-m)\,dt' = \operatorname{sinc}(Bt-m).$$

Hence

$$E[Z_\ell Z_m^*] = \sigma^2 z B^2 \int_{-\infty}^{\infty} \operatorname{sinc}(Bt-\ell)\operatorname{sinc}(Bt-m)\,dt.$$

Using again the orthogonality relation,

$$\int_{-\infty}^{\infty} \operatorname{sinc}(Bt-\ell)\operatorname{sinc}(Bt-m)\,dt = \frac{1}{B}\delta_{\ell m},$$

we obtain

$$E[Z_\ell Z_m^*] = \sigma^2 B z\,\delta_{\ell m}.$$

Therefore, the variables Z_ℓ are uncorrelated. Since they are jointly Gaussian, they are in fact independent. They are also identically distributed, so $Z_\ell \overset{\text{i.i.d.}}{\sim} \mathcal{CN}(0,c)$, with $c = \sigma^2 B z$.

At the receiver, the normalized propagation distance is $z = 1$, so the final variance is $c = \sigma^2 B$.

In physical units, this corresponds to $c = \sigma_0^2 BL$.

Memoryless channel Since $\hat{s}_\ell = s_\ell + Z_\ell$, with Z_ℓ i.i.d. Gaussian, the conditional distribution factorizes:

$$p(\hat{\mathbf{s}}^n \mid \mathbf{s}^n) = \prod_{\ell=1}^{n} p(\hat{s}_\ell \mid s_\ell).$$

Hence the discrete-time channel is memoryless.

Conditional PDF and ML detector For each symbol, the conditional distribution is complex Gaussian:

$$p(\hat{s}_\ell \mid s_\ell) = \frac{1}{\pi c}\exp\left(-\frac{|\hat{s}_\ell - s_\ell|^2}{c}\right).$$

Therefore, the maximum-likelihood detector is

$$\tilde{s}_\ell = \arg\max_{a\in\mathcal{C}} p(\hat{s}_\ell \mid a).$$

Since the exponential is monotone, maximizing the PDF is equivalent to minimizing the squared distance:

$$\tilde{s}_\ell = \arg\min_{a\in\mathcal{C}} |\hat{s}_\ell - a|^2.$$

So the ML detector for this AWGN discrete-time channel is simply the minimum-distance detector.

Decision regions. The decision region of a constellation point $a \in C$ is

$$D(a) = \{u \in \mathbb{C} : |u-a|^2 \le |u-b|^2,\ \forall b \in C\}.$$

These are the Voronoi regions of the constellation points.

For square QAM, the decision boundaries are the midpoints between neighboring constellation points, so detection is nearest-neighbor slicing in the complex plane.

Question 20

Before measuring noisy-channel performance, the complete communication chain was tested with zero noise:

$$\sigma^2 = 0.$$

In this case, the ideal output is

$$\tilde{s} = s, \qquad \hat{b} = b.$$

In the notebook, the verification is reported through:

- $\max |\hat{s} - s|$,
- relative symbol error,
- logical check that $\tilde{s} = s$,
- logical check that $\hat{b} = b$.

```
max |q(t,0) - q_e(t,1)| = 6.6620e-16

[SINC]
max |shat - s| = 0.00096163003828091
relative symbol error = 0.0009379461561788878
stilde == s : True
bhat == b   : True

[RRC]
max |shat - s| = 1.643212638987586e-08
relative symbol error = 1.2851471663734055e-08
stilde == s : True
bhat == b   : True
```

Question 21

Input waveform:

$$q(t,0) = A_1 e^{-\frac{(t+t_0)^2}{2D^2}} + A_2 e^{-\frac{(t-t_0)^2}{2D^2}},$$

with $A_1 = 1$, $A_2 = 2$, $D = 1$.

This signal is the sum of two separated Gaussian pulses centered at $t = -t_0$ and $t = t_0$.

For the linearized fiber channel, the frequency response is

$$\hat{h}(\omega, z) = e^{j\omega^2 z}.$$

Therefore,

$$|\hat{h}(\omega, z)| = 1,$$

which means that the channel is all-pass in frequency: it changes only the phase of each frequency component, not its magnitude.

As a result, the propagated signal $q(t, z)$ may look very different in time, even though the magnitude of its spectrum is unchanged. This is because different frequency components acquire different phase shifts, and after inverse Fourier transform they recombine into a broadened waveform. The guide explicitly states that dispersion is responsible for temporal broadening.

Case 1: $t_0 = 4$**.** At $z = 0$, the two Gaussian pulses are initially separated in time. After propagation to $z = 1$, each pulse broadens because of dispersion. Since the pulses are not very far apart, the broadened tails overlap. Hence dispersion gives rise to inter-symbol interference (ISI) in time.

Case 2: $t_0 = 7$**.** When t_0 is increased to 7, the two pulses are farther apart at the transmitter. After propagation, dispersion still broadens each pulse, but the overlap becomes smaller than in the case $t_0 = 4$. Therefore the ISI is reduced.

Equalization. For the dispersive linear channel, equalization is performed in the frequency domain by multiplying by the inverse response

$$\hat{h}^{-1}(\omega, z) = e^{-j\omega^2 z}.$$

Since this exactly removes the dispersive phase in the absence of noise, the temporal broadening is canceled and the original two-pulse waveform is recovered. Therefore, in the noise-free linear channel, the ISI caused by dispersion is completely canceled with equalization.

Interaction between frequency components. Dispersion does not create interaction between different frequency components in the linear channel. In frequency domain, the channel acts independently on each frequency through the multiplication

$$\hat{q}(\omega, z) = \hat{h}(\omega, z)\hat{q}(\omega, 0).$$

So each frequency component is multiplied by a phase factor, but there is no mixing between distinct frequencies.

In contrast, in time domain the same channel is a convolution

$$q(t, z) = h(t, z) \star q(t, 0),$$

which introduces memory and causes pulse spreading and ISI.

Preferred domains. Modulation is naturally expressed in time domain, since the transmitted waveform is built as a sum of shifted basis pulses.

Equalization is preferred in frequency domain, because dispersion is diagonal there:

$$\hat{q}_e(\omega, z) = \hat{h}^{-1}(\omega, z)\hat{q}(\omega, z).$$

So the channel is complicated in time domain but simple in frequency domain.

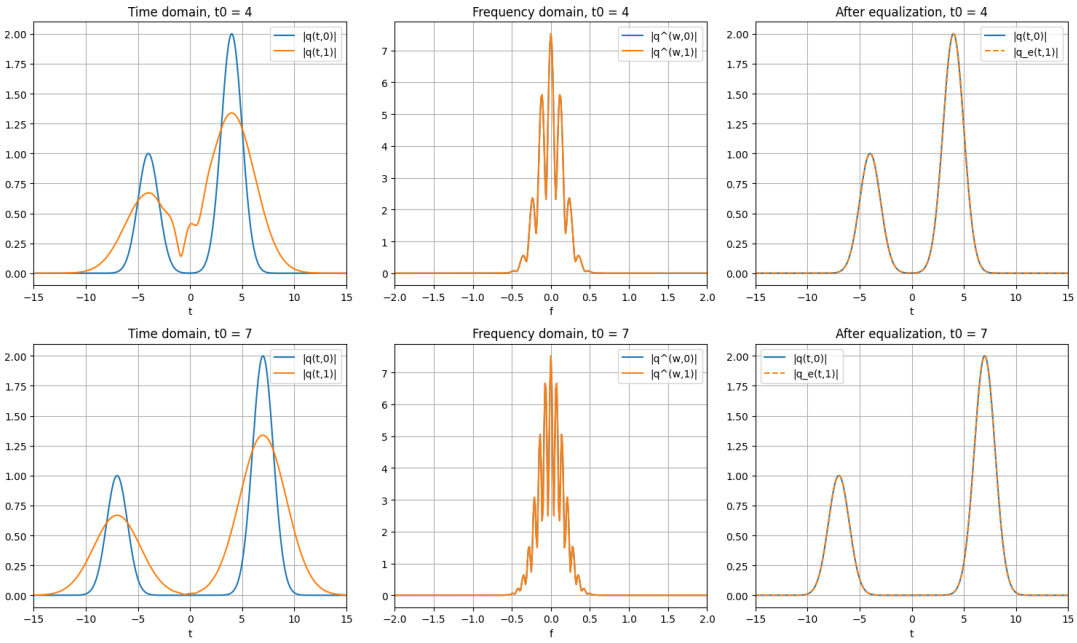


Figure 5: Two-pulse experiment showing temporal spreading and the effect of equalization.

Question 22

The results show the expected behavior: as SNR increases, both BER and SER decrease. The transmitter/receiver constellation plots also show that the received symbol clouds become more concentrated around the ideal constellation points as SNR increases.

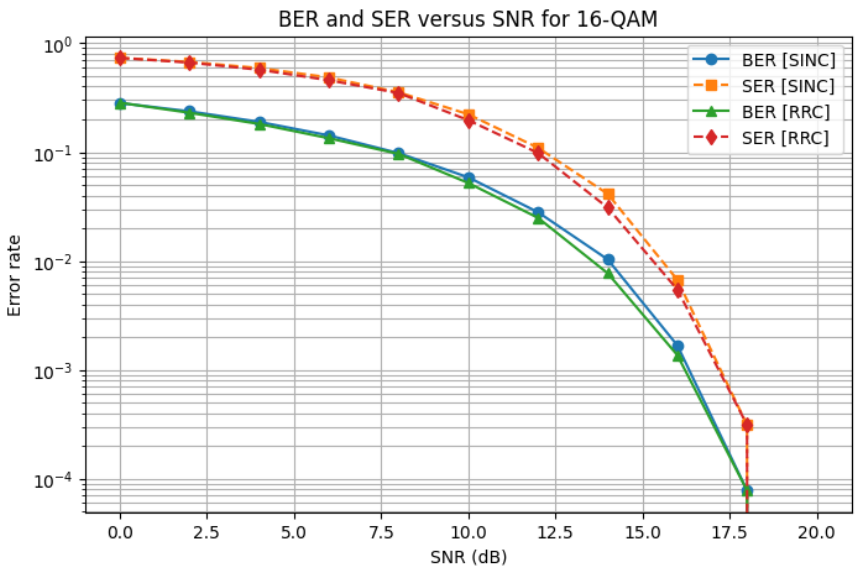


Figure 6: BER and SER versus SNR for the implemented communication system.

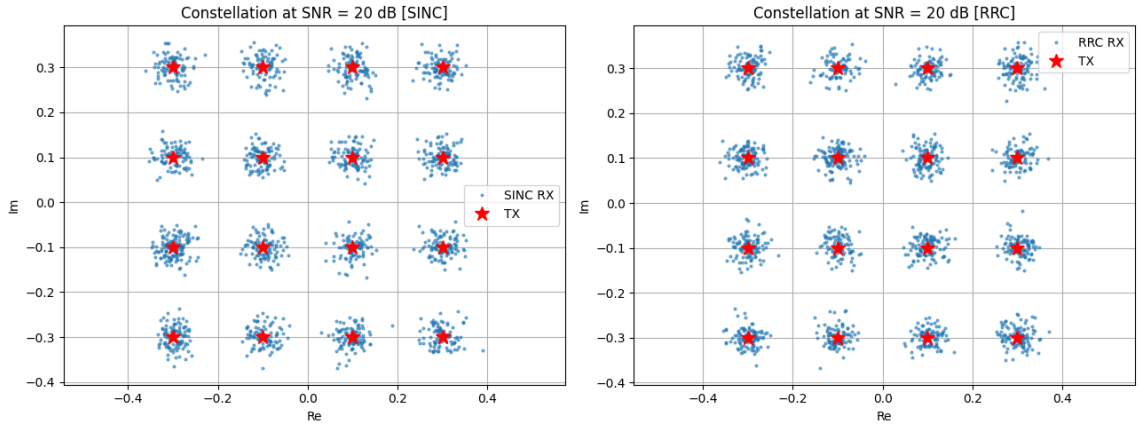


Figure 7: Transmitter/receiver constellation plots at a representative SNR for both pulse shapes.

At this stage, the full linear chain is validated. The remainder of the project then addresses the nonlinear channel and neural equalization.

2 Part II: Neural equalization

2.1 Generative channel model

The following parts address Question 22 and onward.

Noise generation

A circularly symmetric complex Gaussian noise vector

$$Z \sim \mathcal{CN}(0, \sigma^2 I_n)$$

is generated by drawing independent real and imaginary Gaussian components:

$$Z_k = \sqrt{\frac{\sigma^2}{2}}(X_k + jY_k), \qquad X_k, Y_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1).$$

Fast dataset generation with cached pulse matrices

To make the final dataset generation practical, I used a faster modulation/demodulation pipeline based on precomputed pulse matrices. The idea is simple: for a fixed block length, pulse type, bandwidth, and roll-off factor, the sampled shifted basis functions are always the same. Therefore, instead of recomputing the shifted sinc or RRC pulses inside every simulation loop, they are computed once and then reused.

If the shifted pulse basis functions are stacked into a matrix S_m, then modulation can be written in matrix form as

$$q_0 = s\, S_m,$$

where s is the transmitted symbol block and q_0 is the sampled transmitted waveform. In the same way, demodulation can be written as a matched-filter projection

$$\hat{s} = S_m^{(H)} q,$$

where q is the received sampled waveform after propagation and equalization. Thus, both modulation and demodulation reduce to matrix multiplications.

In the implementation, separate cached routines are used for the sinc and RRC pulse shapes. Once the matrices for a given configuration have been generated, later calls reuse the stored result instead of rebuilding the same basis again. This is especially useful for the RRC case, where pulse evaluation is more expensive than for sinc.

The simulation grid is still chosen automatically from the block size and normalized bandwidth, so the underlying communication model is unchanged. The only difference is computational efficiency. In this way, the final fast pipeline remains mathematically equivalent to the direct implementation, but it is much better suited for generating the large number of symbol blocks needed for training and evaluation.

2.2 Training data set

The predictive equalizer is trained on symbol-block pairs of the form

$$(\hat{s}^{(i)}, s^{(i)}),$$

where:

- $s^{(i)}$ is the transmitted symbol block,
- $\hat{s}^{(i)}$ is the received soft symbol block after nonlinear propagation, fast linear equalization, and demodulation.

The guide asks for a training set size between 50,000 and 400,000 symbols. In the final notebook:

$$n_{\text{examples}} = 12500, \qquad n_s = 32,$$

so the total number of symbols is

$$12500 \times 32 = 400000.$$

The final data set is generated at the grading operating point:

$$M = 16, \qquad B = 1.0000 \times 10^1\,\text{GHz}, \qquad L = 1.0000 \times 10^3\,\text{km}, \qquad \text{SNR} = 35\ \text{dB}.$$

2.3 Preprocessing before predictive training

Before training the neural equalizers, the symbol blocks are preprocessed in three steps.

Global phase correction

A single average phase rotation is estimated from the training set and removed:

$$X_{\text{pc}} = X\, e^{-j\theta}.$$

This is a one-time calibration of a deterministic common phase bias. The purpose is to avoid forcing the neural network to spend capacity learning a trivial fixed rotation.

Normalization

The real and imaginary parts are normalized separately using the training-set mean and standard deviation:

$$X_{\text{norm}} = \frac{\Re(X) - \mu_r}{\sigma_r} + j\,\frac{\Im(X) - \mu_i}{\sigma_i}.$$

The same normalization statistics are then applied to validation and test data. This standardization improves numerical stability during training.

Train/validation/test split

The data set is split into

$$80\% \text{ training}, \qquad 10\% \text{ validation}, \qquad 10\% \text{ test}.$$

This allows model selection and early stopping to be performed without using test data.

2.4 Predictive equalizers

Two predictive neural receivers are compared.

Convolutional equalizer

The guide proposes a Conv-FC equalizer. In the final implementation, I used an optimized convolutional alternative: a **dilated residual CNN**. The input to the network is a two-channel sequence containing the real and imaginary parts of the received soft-symbol block. Complex-valued convolution is implemented using paired real convolutions.

The architecture has three main advantages:

- it remains a convolutional many-to-many equalizer as required in spirit by the guide,
- dilation enlarges the receptive field without drastically increasing parameter count,
- residual connections improve optimization and help preserve useful linear structure.

Thus, although it is more optimized than a minimal Conv-FC design, it is still a natural convolutional equalizer for the problem.

BiLSTM-FC equalizer

A BiLSTM-FC equalizer was also implemented. The sequence of received symbols is processed in both forward and backward directions by a bidirectional LSTM, after which fully connected layers produce the output symbol predictions.

This architecture is attractive because the channel has memory. A recurrent model can, in principle, learn dependencies extending over multiple neighboring symbols. In the final results, this model provides a useful comparison point against the convolutional receiver.

2.5 Training procedure

Both neural equalizers are trained using mean-squared error between the predicted and target symbol blocks:

$$\mathcal{L} = \frac{1}{N} \sum_i \|\hat{y}_i - y_i\|^2.$$

The following training choices are used:

- Adam optimizer,
- learning-rate reduction on plateau,
- gradient clipping,
- early stopping based on validation loss.

The baseline against which the neural models are compared is the conventional receiver without neural equalization: the soft symbols are directly detected after phase correction, linear equalization, and demodulation.

2.6 Evaluation metrics

The code reports:

- BER,
- SER,
- mutual information,
- number of trainable parameters.

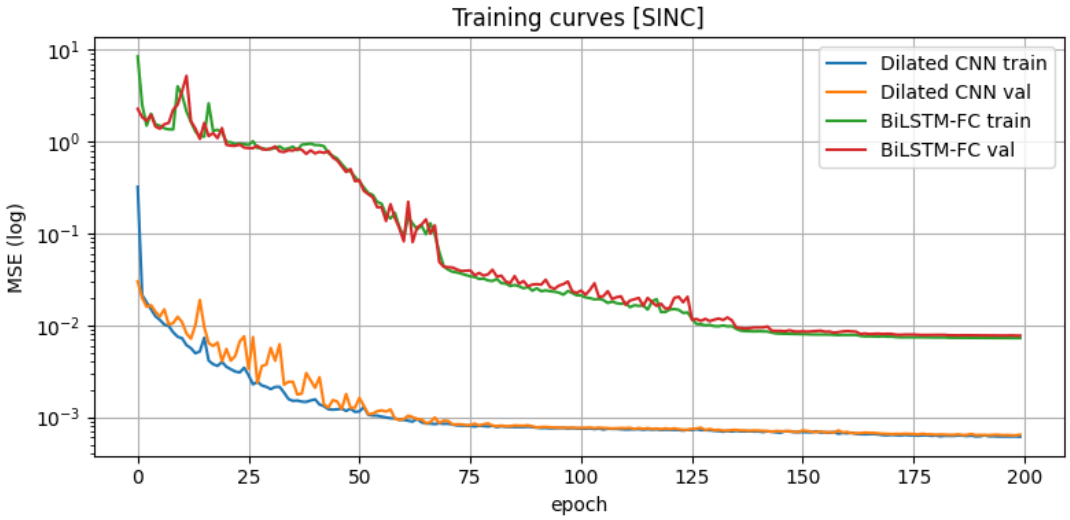
For the final score-oriented comparison, the mutual information is reported using the grading approximation:

$$I_{\text{grading}} = \log_2 M - H(p),$$

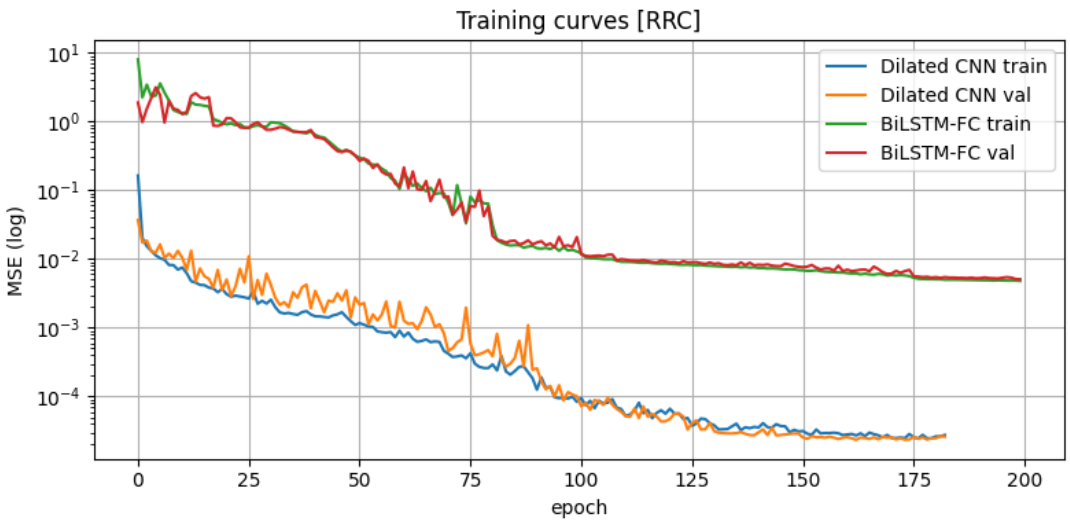
where p is the first row of the empirical symbol-transition matrix. This is the metric required by the grading scheme for the symbols-to-symbol receiver.

2.7 Results

The following figures summarize the model training and the constellation improvements obtained by neural equalization.



(a) Training curves, SINC

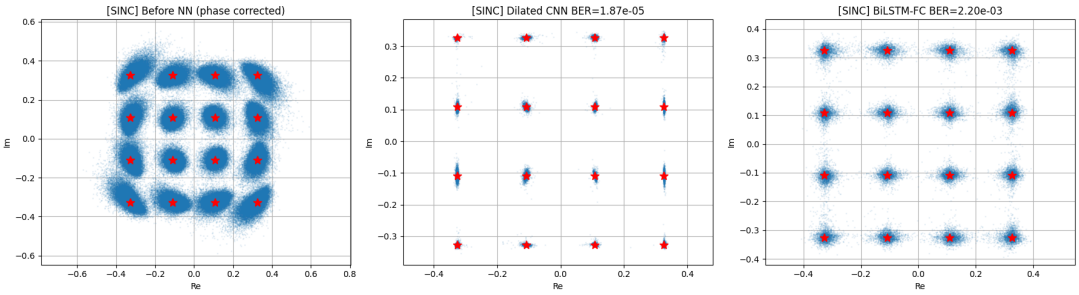


(b) Training curves, RRC

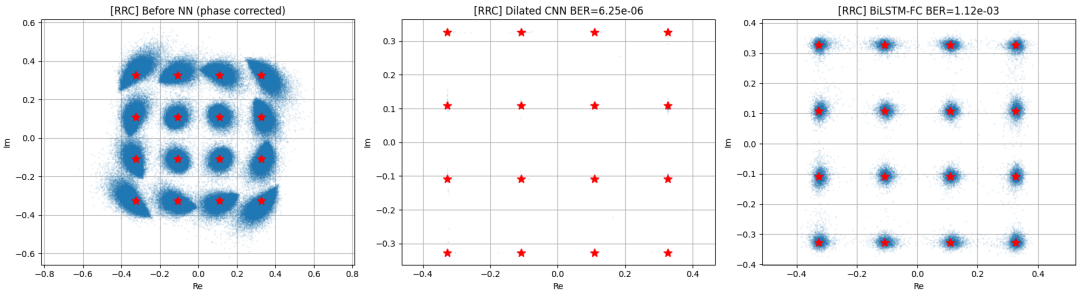
Figure 8: Training and validation loss curves for the predictive neural equalizers.

Table 1: Final comparison results.

Model	BER	SER	MI [bits/sym]	#params
SINC - Linear	4.5531×10^{-3}	–	3.7632	0
SINC - Dilated CNN	1.8750×10^{-5}	7.5000×10^{-5}	4.0000	8542
SINC - BiLSTM-FC	2.2000×10^{-3}	8.5750×10^{-3}	3.9207	51586
RRC - Linear	4.3575×10^{-3}	–	3.7988	0
RRC - Dilated CNN	6.2500×10^{-6}	2.5000×10^{-5}	4.0000	8542
RRC - BiLSTM-FC	1.1187×10^{-3}	4.4500×10^{-3}	3.9761	51586



(a) SINC



(b) RRC

Figure 9: Constellation clouds before and after neural equalization.

Interpretation of the results. Several conclusions follow from the final experiments.

First, the conventional linear receiver is not sufficient for the nonlinear channel. Its BER remains on the order of 10^{-3}, which indicates that significant residual impairment remains after linear equalization alone.

Second, the convolutional neural equalizer achieves the best performance-complexity trade-off in these experiments. It uses relatively few parameters and drives BER to extremely small values for both pulse shapes.

Third, the BiLSTM-FC model is substantially larger but, in this implementation, does not outperform the convolutional receiver. This suggests that for the chosen symbol-block representation and preprocessing, the optimized convolutional model is better matched to the task.

Finally, the RRC pulse gives a slightly better linear baseline than the sinc pulse, but after neural equalization both pulse shapes achieve very strong performance. Thus, in the final score-oriented setting, the dominant factor is the predictive equalizer rather than the small difference between the two pulse choices.